

Concept Review

Camera Model

Why Model a Camera?

Cameras are commonplace items today, often a must both from a technological perspective and user's point of view. Cell phones, cars, gaming consoles, drones etc. all deploy this sensor due to its innate ability to provide vision or sight similar to human eyes. Modeling a camera allows us to understand the limitations of projecting a real 3D world onto an image plane, and how features identified in images could be localized and placed back in the real world. While quite a few types of camera sensors exist with corresponding models, this document will focus on the pinhole camera model.

Camera Extrinsic Properties

A camera's extrinsic definition makes use of a homogeneous transformation matrix to describe a location $\mathbf{t}_j^i$ and rotation $\mathbf{R}_j^i$ with respect to a desired reference frame. For cameras the extrinsic definition is used to convert a point of light $\overrightarrow{\mathbf{P}}_E$ from an environment frame to the camera reference frame. The general form for a homogeneous transformation matrix is,

$$\mathbf{T}_j^i = \begin{bmatrix} \mathbf{R}_j^i & \mathbf{t}_j^i \\ \mathbf{0} & 1 \end{bmatrix} \quad (1)$$

The rotation matrix $\mathbf{R}_j^i$ can be constructed as the combination of 3 successive rotation about the principal axes x, y, z specified in a particular order. $\mathbf{R}_j^i$ defines the relation of frame j with respect to frame i . Rotations about each principal axis: (Spong, 2006)

$$\mathbf{R}_{x,\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix}$$

Rotation about x axis

$$\mathbf{R}_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

Rotation about y axis

$$\mathbf{R}_{z,\psi} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation about z axis

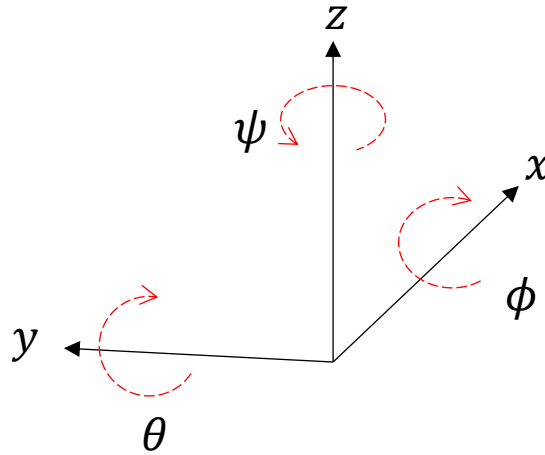


Figure 1: Positive rotation for primary x-y-z axis

With a rotation definition specified the conversion of a point of light $\overrightarrow{\mathbf{P}}_E$ from the environment frame $\mathbf{E}$ to the camera frame $\mathbf{C}$ is,

$$\overrightarrow{\mathbf{P}}_C = \mathbf{T}_E^C \overrightarrow{\mathbf{P}}_E \quad (2)$$

Having defined the point of light in the camera frame the next step is to construct the conversion from a physical point to a pixel in an image.

Camera Intrinsics

Digital cameras are electromechanical devices which use lenses and optical sensors to transform visible light into a digital image. Prior to using a camera for a visual servoing application we need to calculate the intrinsic parameters which describe how the physical point of reference was transformed into a pixel coordinate. All cameras have the following set of equations:

$$K = \begin{bmatrix} f_x & 0 & C_x \\ 0 & f_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Matrix K describes the principal point and the focal point of a camera in pixel coordinates. This matrix can be calculated using the MathWorks™ Camera Calibration tool and a calibration chessboard. Matrix K also depends on the resolution of the images being read by the camera. Perform a new camera calibration when you want to change the resolution of the camera.

Pinhole Camera Model

The pinhole camera model is the first step for converting a physical point of light into an image coordinate. Lens distortion is considered later in the camera model formulation. Point of light $\vec{P}_c$ originates from an optical center called the optical focal point.

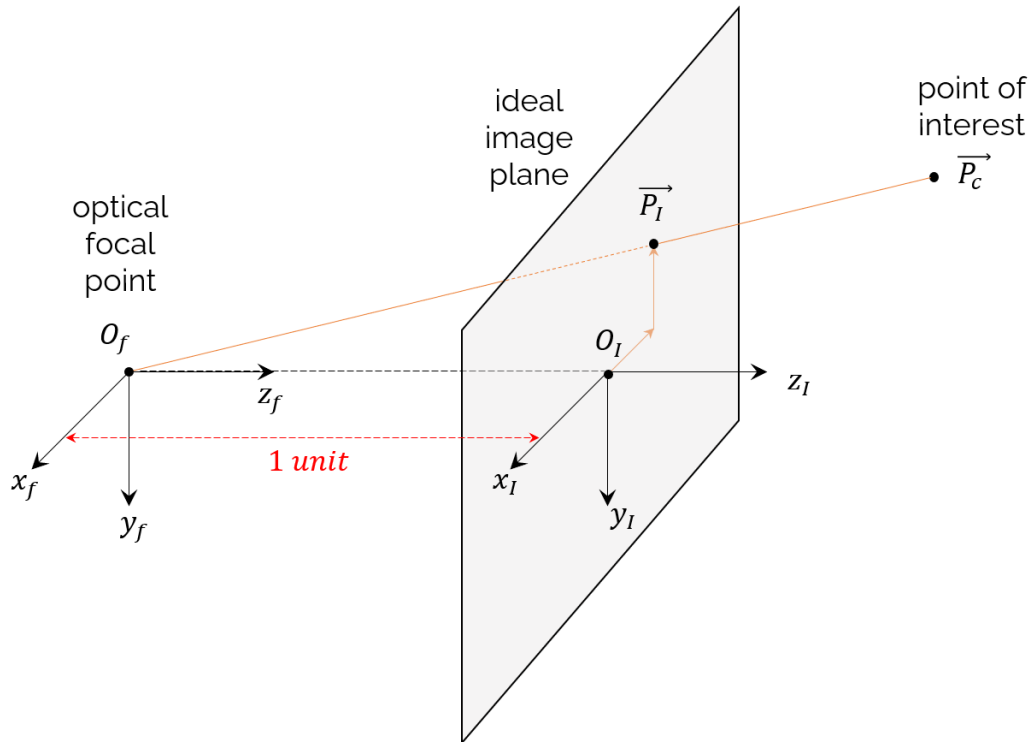


Figure 2: Description of Image Focal point and Ideal image plane. (Waslander, 2021)

Point $\vec{P}_c$ is used to describe the relationship between the optical axis $O_f x_f y_f z_f$ and the ideal image plane originating at $O_I x_I y_I z_I$. Using image projection and similarity triangles point $\vec{P}_c$ can be projected onto the ideal image plane using the following relationship,

$$\vec{P}_I = \begin{bmatrix} Px_I \\ Py_I \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{X}{Z} \\ \frac{Y}{Z} \\ 1 \end{bmatrix} \quad (3)$$

By projection point $\vec{P}_c$ to the ideal image frame there is a change from $\mathbb{R}^3$ to $\mathbb{R}^2$ space. Now in the ideal image plane we consider image distortion due to the type of lens used. The two most common distortion components are,

- Radial distortion (k_{rad})
- Tangential distortion (k_{tan}) .

Radial distortion originates at the center of the image plane and depending on the sign of the leading distortion coefficient, can cause an image to warp inwards in a barrel like fashion, or outwards. The combination of both radial and tangential distortion on the ideal image point $\vec{P}_I$ can be written as,

$$\vec{P}_{Dist} = \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \end{bmatrix} = (1 + k_1 r^2 + k_2 r^4 + K_3 r^6) \begin{bmatrix} Px_I \\ Py_I \end{bmatrix} + \begin{bmatrix} 2\tau_1 Px_I Py_I + \tau_2 (r^2 + 2Px_I^2) \\ 2\tau_2 Px_I Py_I + \tau_1 (r^2 + 2Py_I^2) \end{bmatrix} \quad (4)$$

where $r = \sqrt{Px_I^2 + Py_I^2}$ (Barfoot, 2014). An example of radial distortion is shown in Figure 4.

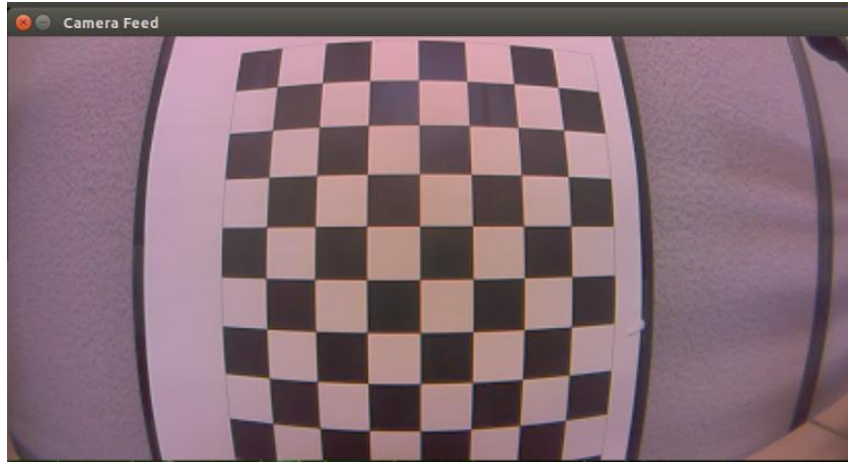


Figure 3: Camera image with barrel distortion.

With the ideal image points modified by the distortion model we now transform from physical coordinates in the ideal image plane to coordinates in the digital image frame. When describing the location of a pixel in an image the following convention is used,

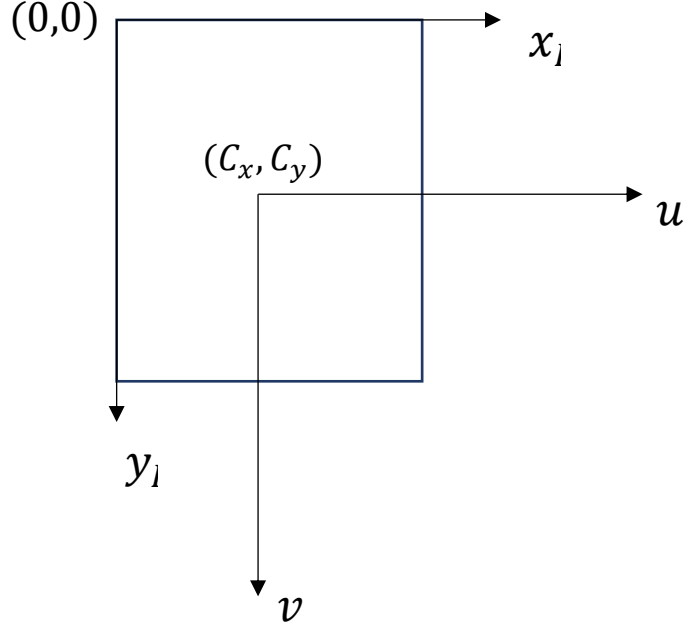


Figure 4: Description of Ideal image to digital image.

With the u axis coinciding with x_l and v axis coinciding with y_l as seen in Figure 4. The transformation between the ideal image plane and the camera image plane is defined as,

$$\begin{aligned} u &= f_x P x_{Dist} + C_x \\ v &= f_y P y_{Dist} + C_y \end{aligned} \quad (5)$$

Where f_x and f_y are the focal lengths along the x and y directions. C_x and C_y are centers of the image. All 4 of these parameters are dependent on image resolution. (Barfoot, 2014)

These steps take us from a 3D point to the image frame. Usually, we need to do the reverse and either calculate 3D information based on information from the camera or rectify distorted image data for a give image processing pipeline. A camera calibration process is used to calculate the camera matrix and distortion parameters.

Summary

Cameras can be modeled using the Camera Obscura (Pinhole camera) model with the inclusion of distortion components. The sequence for transforming a point $\vec{P}_E$ from the global frame to a pixel coordinate (u, v) is,



Camera Frame Transformation

$$\vec{P}_c = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix} \vec{P}_E$$

Image Plane projection

$$\vec{P}_l = \begin{bmatrix} Px_l \\ Py_l \\ 1 \end{bmatrix} = \begin{bmatrix} X/Z \\ Y/Z \\ Z/Z \end{bmatrix} = \frac{\vec{P}_c}{Z}$$

Image distortion

$$\vec{P}_{Dist} = \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \\ 1 \end{bmatrix} = f(\mathbf{k}_{rad}, \mathbf{k}_{tan}, \vec{P}_l)$$

Pixel Image coordinates

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & C_x \\ 0 & f_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \\ 1 \end{bmatrix}$$

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